BEE 4750 Homework 2: Systems Modeling and Simulation

Name: Moho Goswami

ID: mg2297

Due Date

Thursday, 09/25/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to draw a systems diagram and identify the type of a feedback.
- Problem 2 asks you to model contaminant concentrations in a river and use simulation to compare the concentrations to a regulatory standard.
- Problem 3 asks you to explore the implications of an ice-albedo feedback in the Earth's climate system by understanding the equilibria of the modeled system and their stabilities.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```julia
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `c:\Users\mohog\hw2-moho`
   Installed JpegTurbo_jll          v3.1.3+0
   Installed InvertedIndices         v1.3.1
   Installed Accessors              v0.1.42
   Installed WorkerUtilities         v1.6.1
   Installed SentinelArrays          v1.4.8
   Installed InlineStrings           v1.4.5
   Installed Roots                 v2.2.10
   Installed DataFrames             v1.8.0
   Installed WeakRefStrings          v1.4.2
   Installed PooledArrays           v1.4.3
   Installed TableTraits            v1.0.1
   Installed DataValueInterfaces       v1.0.0
   Installed Ghostscript_jll         v9.55.1+0
   Installed ConstructionBase         v1.6.0
   Installed PrettyTables           v3.0.8
   Installed CompositionsBase         v0.1.2
   Installed Crayons               v4.1.1
   Installed InverseFunctions         v0.1.17
   Installed CommonSolve            v0.2.4
   Installed Tables                v1.12.1
   Installed Glib_jll              v2.86.0+0
   Installed Latexify              v0.16.10
   Installed IteratorInterfaceExtensions   v1.0.0
   Installed FilePathsBase           v0.9.24
   Installed StringManipulation        v0.4.1
   Installed CSV                 v0.10.15
Precompiling project...
   1691.4 ms  IteratorInterfaceExtensions
   1125.5 ms  CommonSolve
   1115.4 ms  DataValueInterfaces
   1150.6 ms  WorkerUtilities
   1213.4 ms  InverseFunctions
    946.3 ms  CompositionsBase
    984.2 ms  InvertedIndices
   1185.6 ms  ConstructionBase
   1693.3 ms  InlineStrings
   1346.6 ms  PooledArrays
   3048.8 ms  Crayons
   1958.1 ms  JpegTurbo_jll
   1130.0 ms  TableTraits
   3766.4 ms  SentinelArrays
   2516.7 ms  FilePathsBase
```

```
  2762.8 ms  Glib_jll
  1176.8 ms  LogExpFunctions → LogExpFunctionsInverseFunctionsExt
  1262.0 ms  InverseFunctions → InverseFunctionsDatesExt
  2336.7 ms  InverseFunctions → InverseFunctionsTestExt
  5026.2 ms  StringManipulation
  1130.0 ms  ConstructionBase → ConstructionBaseLinearAlgebraExt
  1217.1 ms  CompositionsBase → CompositionsBaseInverseFunctionsExt
  2172.0 ms  Unitful → InverseFunctionsUnitfulExt
  1380.5 ms  InlineStrings → ParsersExt
  1962.5 ms  Unitful → ConstructionBaseUnitfulExt
  2004.4 ms  Libtiff_jll
  1165.8 ms  FilePathsBase → FilePathsBaseMmapExt
  1969.5 ms  Tables
  3095.1 ms  Ghostscript_jll
  3321.3 ms  FilePathsBase → FilePathsBaseTestExt
  2780.0 ms  Cairo_jll
  2624.0 ms  WeakRefStrings
  3956.6 ms  Qt6Base_jll
  2681.9 ms  HarfBuzz_jll
  6125.7 ms  Accessors
  2399.0 ms  libass_jll
  9145.8 ms  Latexify
  2899.9 ms  Pango_jll
  5894.5 ms  Qt6ShaderTools_jll
  1993.0 ms  Accessors → LinearAlgebraExt
  2143.1 ms  Accessors → TestExt
  2340.8 ms  Accessors → UnitfulExt
  2716.2 ms  Latexify → SparseArraysExt
  3934.4 ms  FFMPEG_jll
  4236.7 ms  UnitfulLatexify
  3128.5 ms  FFMPEG
  3804.6 ms  GR_jll
  8480.5 ms  Qt6Declarative_jll
  9855.2 ms  Roots
  2407.7 ms  Roots → RootsUnitfulExt
 10339.5 ms  GR
 37852.8 ms  CSV
 61530.2 ms  PrettyTables
 78861.6 ms  DataFrames
  6634.0 ms  Latexify → DataFramesExt
122084.5 ms  Plots
  7052.1 ms  Plots → UnitfulExt
 57 dependencies successfully precompiled in 180 seconds. 162 already precompiled.
```

```julia
using Plots
using LaTeXStrings
using CSV
using DataFrames
using Roots
```

Problems (Total: 30 Points)

Problem 1 (5 points)

Draw a systems diagram for the relationship between global mean temperature, atmospheric CO2 concentrations, and ocean CO2 concentrations. What are the signs of the interactions between these components and why? What does this suggest about the overall feedback between temperature and the ocean carbon cycle?

System components * Global mean temperature * Atmospheric CO2 concentrations * Ocean CO2 concentrations

Interconnections * Atmospheric CO2 concentrations INCREASES global mean temperature * Ocean CO2 concentrations DECREASES atmospheric CO2 concentrations * Global mean temperature DECREASES Ocean CO2 concentrations

The above relationships, as described in Woodard et al. (2019), reflect how changes in Earth's carbon balance relate to mean temperature. Increasing the atmospheric CO2 concentrations will trap more carbon which stores more heat, thus increasing temperature. In contrast, the Earth's oceans act as a carbon sink and thus when ocean CO2 concentrations increase, the amount of CO2 in the immediate atmosphere is expected to decrease. However, an increase in global mean temperature, including an increase in ocean temperatures, can decrease the natural ability of oceans to dissolve CO2 due to increased stratification and decreased solubility in accordance with Henry's Law.

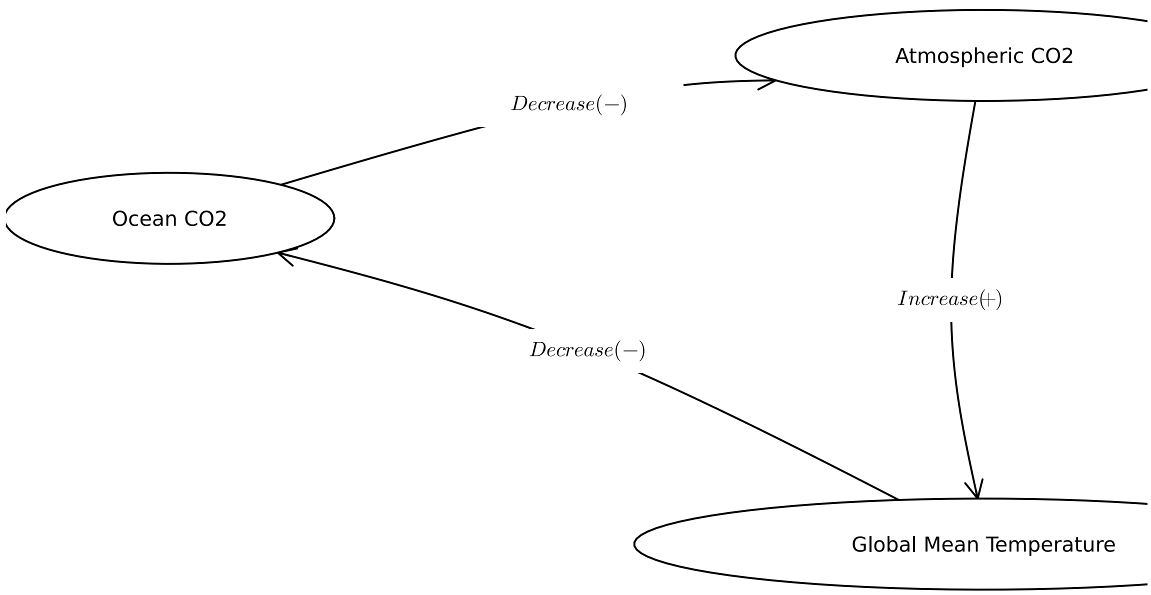
```julia
Pkg.add("GraphRecipes")
Pkg.add("Plots")
using GraphRecipes
using Plots
```

```
  Resolving package versions...
 No Changes to `C:\Users\mohog\hw2-moho\Project.toml`
 No Changes to `C:\Users\mohog\hw2-moho\Manifest.toml`
  Resolving package versions...
 No Changes to `C:\Users\mohog\hw2-moho\Project.toml`
 No Changes to `C:\Users\mohog\hw2-moho\Manifest.toml`
```

```
A = [0 0 1;
     1 0 0;
     0 1 0]

names = ["Atmospheric CO2", "Ocean CO2", "Global Mean Temperature"]
edge_labels = Dict((2, 1) => L"Decrease (-)", (1,3) => L"Increase (+)", (3, 2) => L"Decrease
shapes=[:ellipse :ellipse :ellipse]
xpos = [-0.25, -1.5, -0.25]
ypos = [0.75, 0.5, 0]

p = graphplot(A, names=names,edgelabel=edge_labels, markersize=0.12, markershapes=shapes, ma
display(p)
```



Considering the overall feedback between temperature and the ocean carbon cycle, this graph suggests that increases in global mean temperature due to increases in atmospheric carbon dioxide levels from human activities such as burning fossil fuels can disrupt the extent of the ocean's natural carbon cycle by reducing its carbon uptake capabilities. In accordance with Henry's Law, which indicates that increasing temperature can decrease the ability of liquids to absorb gas, this means that the oceans have a diminished ability to dissolve carbon dioxide.

Tip

Think about Henry's law for CO2.

Problem 2 (15 points)

A river which flows at 10 km/d is receiving discharges of wastewater contaminated with CRUD from two sources which are 15 km apart, as shown in the Figure below. CRUD decays exponentially in the river at a rate of 0.36 d^{-1} and is deposited by the atmosphere along the river at a rate of 54 kg/km/d.

Schematic of the river system in Problem 2

Problem 2.1

Draw a systems diagram with the relevant control volume(s) denoted and any relevant in/outflows between these boxes. How did you decide how many boxes were needed?

- Control volumes: As depicted in the figure, there are 2 inflows coming from each source with different flow rates and CRUD discharge amounts. Thus, 2 boxes will be needed to isolate the interaction of each of these wastewater discharge flows with the river itself.

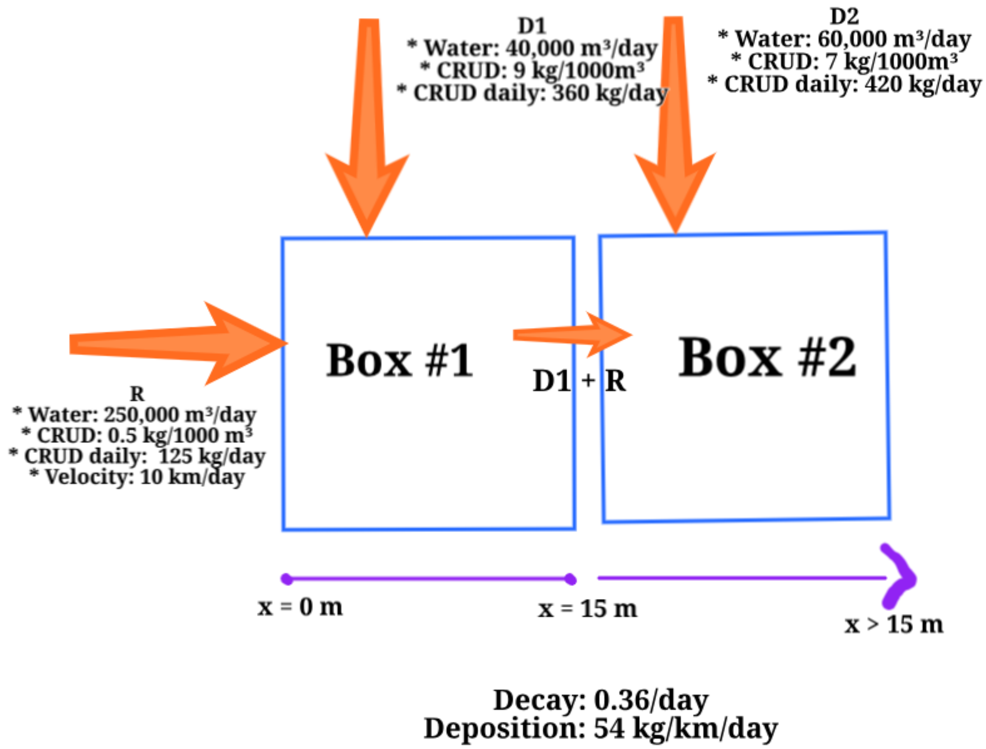


Figure 1: Screenshot 2025-09-22 145737.png

Problem 2.2

Develop a model for the concentration of CRUD downriver by formulating and solving the appropriate differential equation(s) analytically.

> **Tip**
>
> Formulate your model in terms of distance downriver, rather than leaving it in terms of time from discharge.

```
# Defining variables
# k = 0.36/day
# v = velocity, km/day
## k/v = decay per km (REMOVED)
# S = 54 kg/km/day (ADDED)
# C = concentration
# t = time (days)
# x = distance (km)
```

```
# Q = flow rate (km/day)

# dC/dt = -k*C(t) ACCOUNTS FOR DECAY, NEED DEPOSITION
# Converting to dC/dx

# Box 1 Inflow = R + D1
# Outflow from Box 1: QrCr + Qd1Cd1
# Box 2 Inflow = D2 + R + D1
# Outflow from Box 2: Qd2Cd2 + Q(r+d1)C(r+d1)

# INCREASE FROM DEPOSITION, S/Q, final units = kg

# Model
# dC/dX = (-k/v) * C + S/Q
```

Discretizing above differential equation * dC/dX = (-k/v) * C + S/Q

Applying Euler approximation * df/dt ~= (f(t + Δt) - f(t)) / Δt * dC/dX ~= (C(x + Δx) - C(x)) / Δx * (C(x + Δx) - C(x)) / Δx = (-k/v) * C + S/Q

- C(x + Δx) = C(x) + Δx * ((-k/v) * C + S/Q)

Problem 2.3

Determine if the system in compliance with a regulatory limit of 2.3 kg/(1000m^3). You can do this analytically or computationally.

Using * C(x + Δx) = C(x) + Δx * ((-k/v) * C + S/Q)

Problem 3 (10 points)

In class, we discussed the ice-albedo feedback and its possible influence on melting the hypothesized "Snowball Earth". In this problem, we'll introduce a simple model of the Earth's energy balance with includes this feedback and examine the stability of the climate.

This simple model of the energy balance (averaged over the entire planet) is:

$$\underbrace{C\frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1-\alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A-BT)}_{\text{outgoing radiation}} + \underbrace{a\ln\left(\frac{[CO_2]}{[CO_2]_{PI}}\right)}_{\text{greenhouse effect}}, \quad (1)$$

where:

- T is the Earth's global mean temperature (in °C);
- C is the heat capacity of the atmosphere and shallow ocean, taken to be 51 J/m^2/°C;
- α is the planetary albedo, or the fraction of incoming radiation reflected by the Earth, which has a present-day value of approximately 0.3;
- S is the solar constant, or the amount of solar radiation received by the Earth averaged over area, which has a present-day value of 1368 W/m^2 but during the Neoproteorozoic Era had a value of 1272 W/m^2 (this is divided by four because the Earth is a sphere but S is the radiation captured by a disc with radius equal to that of the Earth);
- A and B are coefficients from the linearization of outgoing radiation physics, and have corresponding values $B = -1.3$ W/m^2/°C (estimated from a number of lines of evidence about the sensitivity of outgoing radiation to temperature; this is negative due to a sign convention about the direction of incoming vs. outgoing radiation) and $A = 221.2$ W/m^2 (estimated by assuming the pre-industrial temperature of 14°C was stable without anthropogenic greenhouse gas emissions).

We will ignore the greenhouse effect term as we are considering the Earth system well before humans were around.

Problem 3.1

Discretize the climate model (Equation 1) using forward Euler integration and a time step of $\Delta t = 1$ yr.

```
# this function computes the increment by which the concentration
# is updated at each step
function box_simulate_timestep(T, x, S, C, A, B)
    return ((1-x) * S)/(4*C) - (A-(B*T))/C
end

# this function loops over the timesteps to simulate
# the concentration series
function climate_simulate(x, S, C, A, B, T , Tf, Δt)
    # initialize T storage
    # for code simplicity we make the array length T+1 so
    # index 1 is T
    steps = Int64(Tf / Δt)
    Ts = zeros(steps + 1)
    Ts[1] = T
    for t = 1:steps
        Ts[t+1] = Ts[t] +
            Δt * box_simulate_timestep(Ts[t], x, S, C, A, B)
    end
    # the first element of T is the initial condition
```

```
    return Ts
end
Δt = 1.0 #Time step of 1 year
Tf = 10 #End-time
T  = 14 #Pre-industrial temperature
C = 51 #J/m^2/degrees C
x = 0.3 #Planetary albedo alpha
S = 1272 #W/m^2 #Neoproterozoic value
A = 221.2 #W/m^2
B = -1.3 #W/m^2/degrees C
Ts = climate_simulate(x, S, C, A, B, T , Tf, Δt)
```

```
11-element Vector{Float64}:
 14.0
 13.670588235294119
 13.349573241061131
 13.03674098197526
 12.731882878513145
 12.434795667884378
 12.145281268506933
 11.863146647937148
 11.5882036941662
 11.320269090197257
 11.05916419181968
```

Problem 3.2

Rather than assuming a constant value for the albedo α, we will represent the ice-albedo feedback by letting α depend on T:

$$\alpha(T) = \begin{cases} \alpha_i & \text{if } T \leq -10^\circ\text{C} \\ \alpha_i + (\alpha_0 - \alpha_i)((T+10)/20) & \text{if } -10^\circ\text{C} \leq T \leq 10^\circ\text{C} \\ \alpha_0 & \text{if } T \geq 10^\circ\text{C}. \end{cases}$$

Let $\alpha_i = 0.5$ and $\alpha_0 = 0.3$. Simulate the simple climate model using the Neoprotereozoic value of S with temperature-varying albedo for initial values of T spanning $-60^\circ\text{C} \leq T_0 \leq 30^\circ\text{C}$ over a period of 200 years. Plot the temperature trajectories. How many equilibria are there and what are their stabilities?

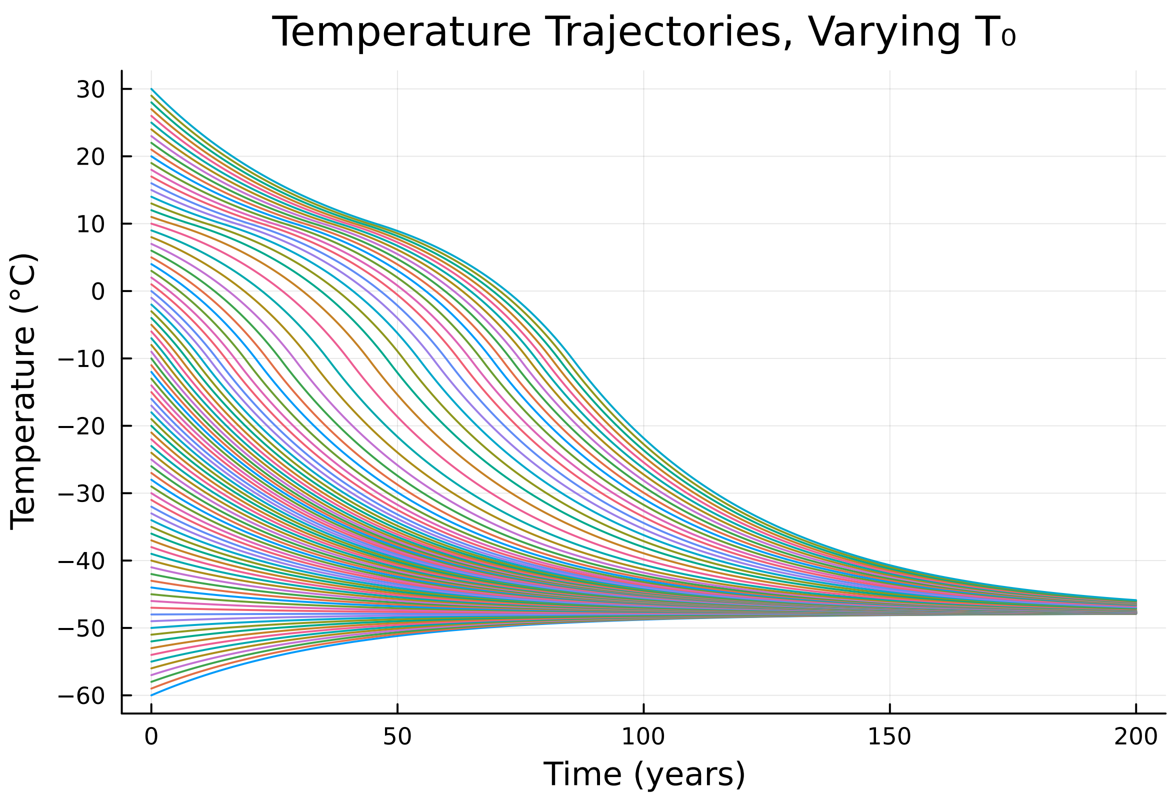
```
# this function computes the increment by which the concentration
# is updated at each step
function box_simulate_timestep(T, xi, x , S, C, A, B)
    if T <= -10
         x = xi
    elseif -10 <= T <= 10
        x = xi + (x -xi)*((T+10)/20)
    elseif T >= 10
         x = x
    end

    return ((1-x) * S)/(4*C) - (A-(B*T))/C
end
# this function loops over the timesteps to simulate
# the concentration series
function climate_simulate(xi, x , S, C, A, B, T , Tf, Δt)
    # initialize T storage
    # for code simplicity we make the array length T+1 so
    # index 1 is T
    steps = Int64(Tf / Δt)
    Ts = zeros(steps + 1)
    Ts[1] = T
    for t = 1:steps
        Ts[t+1] = Ts[t] +
            Δt * box_simulate_timestep(Ts[t], xi, x , S, C, A, B)
    end
    # the first element of T is the initial condition
    return Ts
end
Δt = 1.0 #Time step of 1 year
Tf = 200 #End-time of 200 years
C = 51 #J/m^2/degrees C
x = 0.3 #Planetary albedo alpha
xi = 0.5
x  = 0.3
S = 1272 #W/m^2 #Neoproterozoic value
A = 221.2 #W/m^2
B = -1.3 #W/m^2/degrees C

# setting temperature range from -60 C to 30 C
T _values = -60.0:1.0:30.0
```

```
Ts = Dict{Float64, Vector{Float64}}()
for T  in T _values
    Ts[T ] = climate_simulate(xi, x , S, C, A, B, T , Tf, Δt)
end

plot(yticks = -60:10:30)
for (T , T _values) in sort(collect(Ts))
    plot!(0:Δt:Tf, T _values, label="", lw=1)
end
xlabel!("Time (years)")
ylabel!("Temperature (°C)")
title!("Temperature Trajectories, Varying T ")
```



From the graph, it is visible that there is 1 equilibrium at approximately -47 degrees Celsius. This equilibrium is stable: the varying temperature values all converge to the same equilibrium without significant fluctuations.

Problem 3.3

One might hypothesize that one cause for the transition from Snowball Earth (the stable frozen equilibrium from Problem 3.2) was an increasing amount of incoming solar radiation (as expressed by an increase in S). Examine this hypothesis using our simple model. Is this factor enough to cause the planet to warm to the pre-industrial temperature of 14°C?

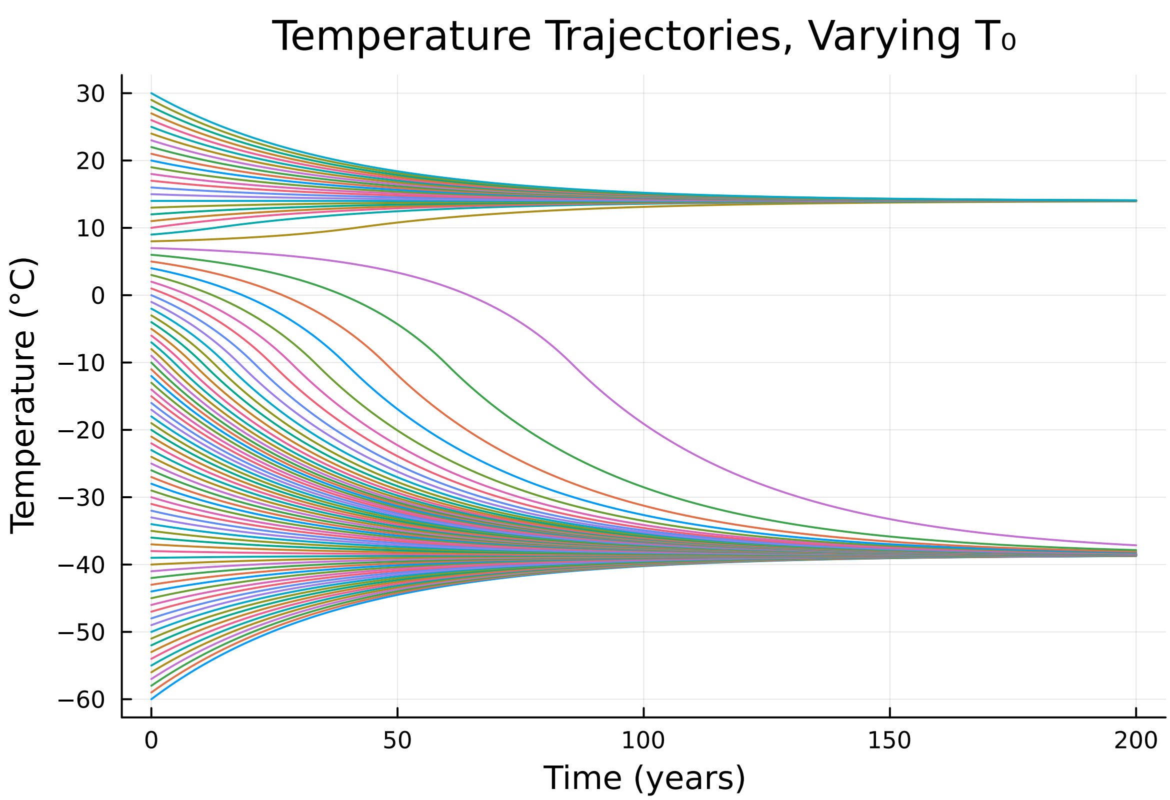
```
# this function computes the increment by which the concentration
# is updated at each step
function box_simulate_timestep(T, xi, x , S, C, A, B)
    if T <= -10
         x = xi
    elseif -10 <= T <= 10
        x = xi + (x -xi)*((T+10)/20)
    elseif T >= 10
         x = x
    end

    return ((1-x) * S)/(4*C) - (A-(B*T))/C
end
# this function loops over the timesteps to simulate
# the concentration series
function climate_simulate(xi, x , S, C, A, B, T , Tf, Δt)
    # initialize T storage
    # for code simplicity we make the array length T+1 so
    # index 1 is T
    steps = Int64(Tf / Δt)
    Ts = zeros(steps + 1)
    Ts[1] = T
    for t = 1:steps
        Ts[t+1] = Ts[t] +
            Δt * box_simulate_timestep(Ts[t], xi, x , S, C, A, B)
    end
    # the first element of T is the initial condition
    return Ts
end
Δt = 1.0 #Time step of 1 year
Tf = 200 #End-time of 200 years
C = 51 #J/m^2/degrees C
x = 0.3 #Planetary albedo alpha
xi = 0.5
x  = 0.3
S = 1368 #W/m^2 #MODERN value
A = 221.2 #W/m^2
B = -1.3 #W/m^2/degrees C
```

```
# setting temperature range from -60 C to 30 C
T _values = -60.0:1.0:30.0

Ts = Dict{Float64, Vector{Float64}}()
for T  in T _values
    Ts[T ] = climate_simulate(xi, x , S, C, A, B, T , Tf, Δt)
end

plot(yticks = -60:10:30)
for (T , T _values) in sort(collect(Ts))
    plot!(0:Δt:Tf, T _values, label="", lw=1)
end
xlabel!("Time (years)")
ylabel!("Temperature (°C)")
title!("Temperature Trajectories, Varying T ")
```



After copying and pasting the same code that was used in 3.2 and only increasing the value of S from 1272 to 1368 W/m^2, the graph reaches a second equilibrium at approximately T = 14 degrees Celsius, given that the initial temperature was at least 8 degrees Celsius. Although a higher equilibrium can eventually be reached, changing the amount of solar radiation does not provide the initial boost needed to increase the temperature to at least 8 degrees Celsiuc for the equilibrium shift to occur. As such, while increased solar radiation could plausibly have been a contributing factor to temperature increases over time, based on this graph it would not have been impactful enough to be the sole reason for the transition from "Snowball Earth" to a pre-industrial planetary temperature of 14 degrees C.

References

List any external references consulted, including classmates.

HW 1 Code * Generating graph for #1

Lecture 6 slides * 3.1 Forward Euler code

https://discourse.julialang.org/t/how-to-work-with-intervals-values-in-julia/6433/5 * 3.2

Lecture 5 slides * 3.2